

DATA VISUALIZATION • COM-480 • EPFL 2026

CineScope

What makes a movie successful?

A larger budget. The right genre. Audience love. A sequel. A trusted director. Cinema sells these promises before the lights go down. CineScope checks what survives after opening night.

Hamza Barrada • 327986 Amer Lakrami • 344911
Ziad Chentouf • 344912 Salma El Yadouni • 340859

ASTRA CINE - STARDUST DATA LOG

ASTRA CINE
STARDUST DATA PROJECT

MILESTONE 3 PROCESS BOOK • TEAM XLB

THE PROJECT

Data, navigation and design language

Dataset and data pipeline

CineScope is built from The Movies Dataset (Kaggle/TMDb), enriched with MovieLens audience ratings and TMDb collection metadata. The raw source contains 45,697 cleaned rows, but missing and implausible financial records make ROI unstable without a reduction step. The preprocessing pipeline requires positive recorded revenue, a minimum production budget of \$10,000, and a unique TMDb identifier. The \$10,000 floor was chosen after examining the tail of the budget distribution: it removes entries clearly caused by data-entry errors while retaining genuine low-budget releases such as *Paranormal Activity* (\$15,000 recorded budget).

Stage	Records
Cleaned source rows	45,697
Positive budget and revenue	5,417
Removed below \$10,000 floor	65
Removed duplicate IDs	35
Final eligible financial films	5,317
Rating-qualified (50+ ratings)	4,607
Eligible franchise collections	309
Eligible sequel comparisons	573
Repeat directors (≥6 films)	181

Definitions

ROI = gross / production budget. 3.0× means gross is three times recorded budget. **Profitable** is shorthand for $ROI \geq 1\times$ which means it means budget recovery, not accounting profit, because marketing, exhibitor share and ancillary revenue are unavailable. Budget tiers: Micro < \$5M; Low \$5–20M; Mid \$20–60M; Blockbuster > \$60M.

Why actors are excluded as a ranking factor

Top-billed cast is shown in individual film dossiers as contextual evidence. We deliberately chose not to rank performers as independent success drivers. An actor’s appearance co-occurs with budget level, franchise membership, director choice and role context. Attributing box-office receipts to an isolated actor influence would claim more than this observational record can defensibly establish, and could generate a misleading “actor leaderboard” that conflates correlation with cause.

Navigation and shared controls

The site opens with a dark cinematic hero and six chapter-entry cards. Each card links directly to one of the six analytical scenes. The fixed navigation bar allows jumping to any chapter; the Back to Top button returns to the opening frame.

The **genre pills** (All, Action, Drama, Comedy, Horror, Animation, Thriller, Sci-Fi, Other) and the **decade slider** (All, 1970s–2010s) form the active analytical cut. Selecting a genre or decade in one view does not require re-selection in another: the state persists across all relevant visualisations simultaneously, coordinated through `js/state.js`. This means a reader who wants to test the Horror hypothesis can select Horror once and carry that cut through all six chapters.

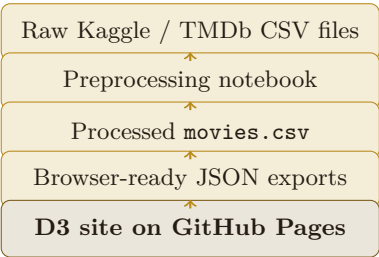
Each view has **local controls** in addition to the shared cut: the Dossier Board has a Revenue/ROI toggle and title search; the Flow has an outcome-mode toggle and hover isolation; the Matrix has a metric toggle and cell drill-down; Dynasties has a metric toggle and collection search; Applause has a Revenue/ROI toggle and band selection; Directors has a minimum-film threshold and rank-metric toggle.

Design

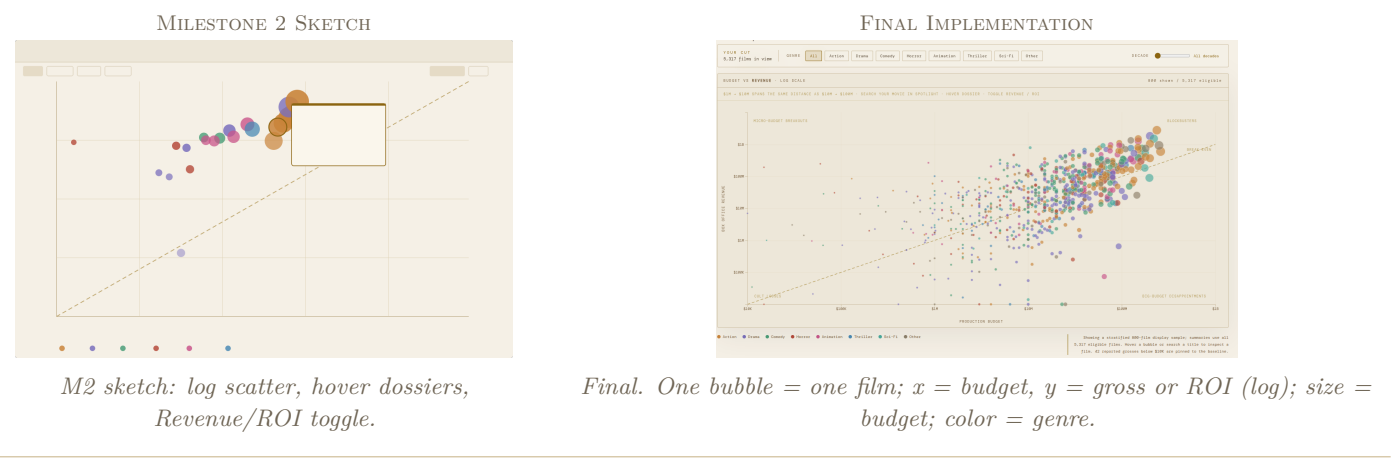
The default **parchment light room** uses warm off-white backgrounds and high-contrast typography, matching the Milestone 2 sketch aesthetic and ensuring readability at all zoom levels. The **dark room** (toggled via the header button) provides a projection-screen palette suitable for screencast recording. It swaps CSS custom properties only; chart modules and analytical logic are identical in both modes.

Typography uses proportional serif text for editorial prose and small caps for labels, echoing classic cinema credits. The gold accent color (`#B98A1C`) is drawn from the cover artwork and applied consistently to headings, active controls and key values. The closing credits screen uses the dark-room palette deliberately, ending the story like a projected title card rather than a data dashboard.

Reproducible pipeline



The Dossier Board: individual films, scale and efficiency



From sketch to final

The prototype already had the right structure: one film per point, budget on x and a commercial outcome on y. The final version adds the shared genre/decade filter bar, title search (Spotlight), richer hover dossiers with poster art and full film context, the log-scale annotation marquee, labelled quadrant regions and the break-even diagonal.

The most consequential implementation choice was **display sampling**. The default cut contains 5,317 films; plotting all of them produces an unreadable mass of overlapping bubbles. We use a deterministic stratified sample of at most 800 marks: films are stratified by genre, budget tier, ROI outcome and decade, and the same seed always produces the same visible selection. Crucially, every written summary — median ROI, recovery rate, correlations — is computed from all 5,317 eligible films, not the 800 displayed. This is stated explicitly in the chart footnote.

Design decisions

We chose **logarithmic scales** for both axes because budget and gross span five orders of magnitude (\$10K to \$3B). On a linear axis, 99 % of films collapse into the bottom-left corner. Log-log also gives the break-even line a clean diagonal: every point equidistant from the diagonal represents the same ROI multiple.

Results by budget tier and genre (default, all decades)

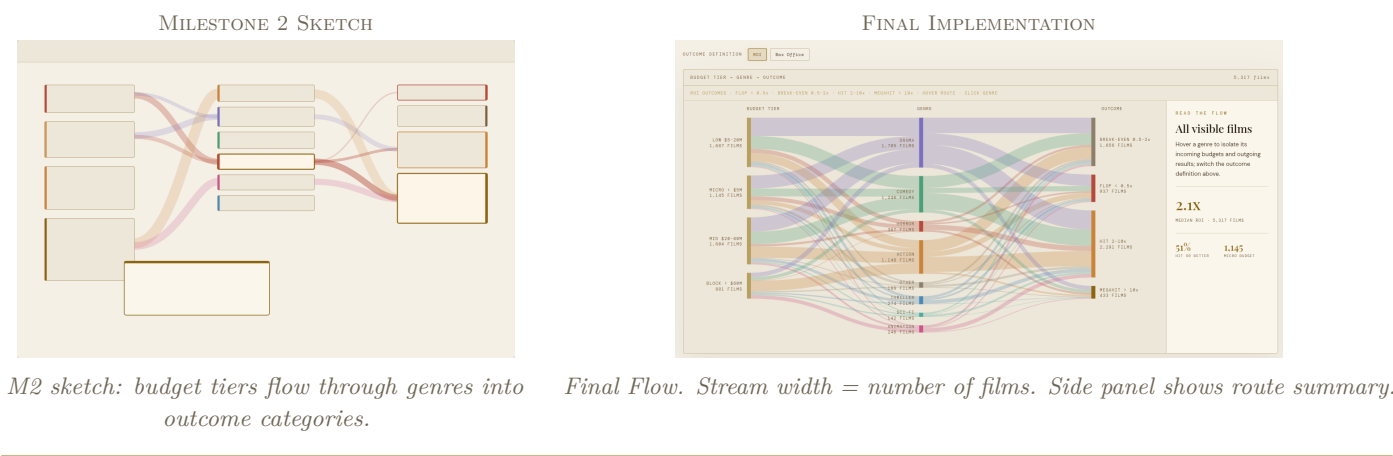
Tier	n	Med. ROI	Recover	Med. gross
Micro <\$5M	1,145	2.98×	74.5 %	\$4.3M
Low \$5–20M	1,687	1.81×	64.0 %	\$18.5M
Mid \$20–60M	1,604	1.73×	67.1 %	\$55.2M
Block >\$60M	881	2.24×	82.2 %	\$211.8M

Micro and Blockbuster tiers both outperform Low and Mid on median ROI, but Blockbuster achieves this with 50× the median gross, while Micro achieves it on a fraction of the budget. Hit or Megahit share (ROI ≥ 2×): Micro 60.9 %; Low 47.4 %; Mid 45.4 %; Blockbuster 56.6 %.

Genre	n	Med. ROI	Med. gross
Horror	367	3.01×	\$27.2M
Animation	246	2.73×	\$139.9M
Comedy	1,246	2.19×	\$32.9M
Sci-Fi	142	2.19×	\$62.6M
Action	1,148	2.00×	\$52.7M
Thriller	274	2.00×	\$32.1M
Drama	1,705	1.74×	\$18.7M

Horror leads by ROI; Animation leads by gross. These two metrics select different genre winners .

The Flow: budget, genre and outcome routes



From sketch to final

The sketch defined a three-stage Sankey. The final view adds explicit outcome threshold definitions, a dynamic route-summary panel and a clickable genre node that transfers the selection into the shared cut. The biggest addition over the sketch is the **ROI versus Box Office outcome toggle**: it classifies the same 5,317 films by two definitions of success simultaneously, making the choice explicit rather than baked silently into the chart.

ROI outcomes: Flop < 0.5×; Break-even 0.5–2×; Hit 2–10×; Megahit > 10×. Box Office outcomes: Under \$10M; \$10M–\$100M; \$100M–\$500M; Over \$500M.

Design decisions

A thick stream means many films took that route; it says nothing about whether they profited. We considered encoding width by median ROI or profitable share, but this created a misleading chart where thin streams (few films, high ROI) appeared insignificant compared with thick streams (many films, lower ROI). The frequency encoding is honest; the side panel delivers the quality metric.

The “**Break-even**” label was a deliberate choice over “Profitable”. Break-even (0.5–2× ROI) honestly communicates that many films in this band are covering costs but not generating returns. “Profitable” would imply accounting profit, which requires marketing, distribution and ancillary data we do not have.

What the data shows

In ROI mode, micro-budget films send 61 % of titles to Hit or Megahit outcomes; mid-budget films send 45 %. Horror leads displayed genres by median ROI at 3.0×. While in Box Office mode Animation dominates at \$139.9M median gross and for Horror, the ROI leader, drops back significantly.

So the Flow makes that definitional choice observable: a genre can be the best route by efficiency while being an average route by scale. The Matrix then shows where within each genre the efficiency advantage concentrates.

Challenge: cross-filter coordination

Clicking a genre in the Flow updates the shared genre cut, which re-renders the Dossier Board, Matrix, Dynasties and Applause views. Making this feel instantaneous required pre-computing all filter results at page load and storing them in `js/state.js`. The decade filter interacts with the genre filter multiplicatively, so all 45 combinations (9 genres × 5 decades) are prepared in the JSON exports. Early prototype versions re-filtered on every interaction, causing noticeable lag on the Matrix violin recomputation. Pre-computing eliminated this entirely.

Flow outcome breakdown ROI mode and Box Office mode (default)

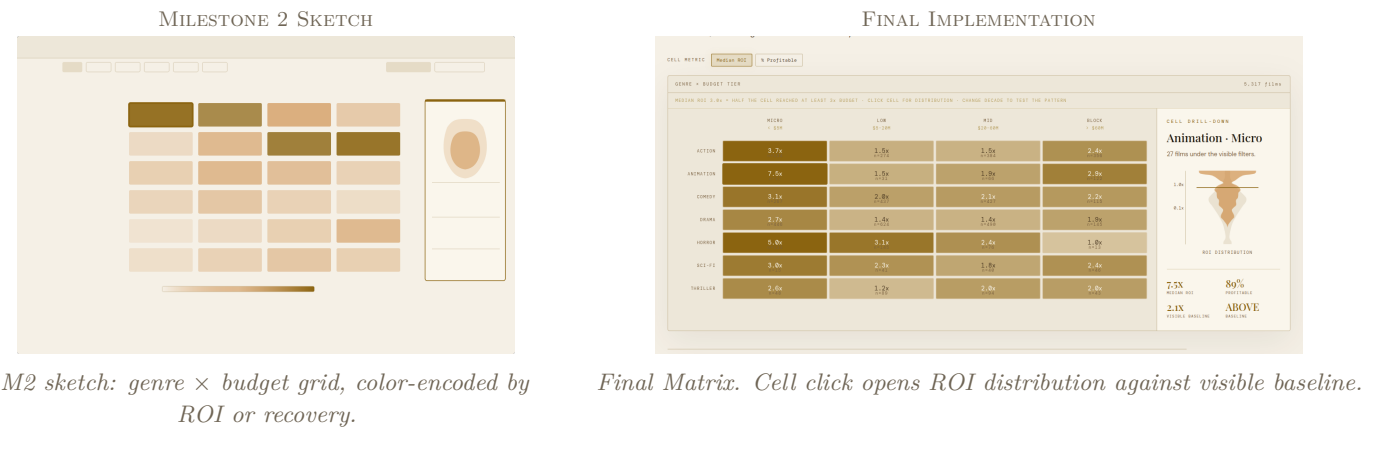
ROI outcome	Films	Share
Flop <0.5×	937	17.6 %
Break-even 0.5–2×	1,656	31.1 %
Hit 2–10×	2,291	43.1 %
Megahit >10×	433	8.1 %

51 % of all 5,317 eligible films reach Hit or better (ROI ≥ 2×). The Horror route sends the highest share to Hit/Megahit in ROI mode; Drama sends the largest volume to Break-even.

Box Office tier	Films	Share
Under \$10M	904	17.0 %
\$10M–\$100M	2,003	37.7 %
\$100M–\$500M	1,670	31.4 %
Over \$500M	740	13.9 %

In Box Office mode the Animation stream narrows but flows almost entirely into the top two tiers; the Micro-budget stream, dominant in ROI Hit outcomes, almost vanishes above \$100M.

The Profitability Matrix: genre-budget fit



M2 sketch: genre × budget grid, color-encoded by ROI or recovery.

Final Matrix. Cell click opens ROI distribution against visible baseline.

From sketch to final

The sketch relied entirely on color to communicate differences. The final version adds exact median ROI values inside every cell, visible sample sizes (*n*), a Median ROI / % Profitable toggle, and a selected-cell distribution panel showing a violin plot against the full-cut baseline.

The violin panel answers a question that a heatmap cell cannot: is a 7.5× median ROI the result of a consistently strong strategy, or of one outlier title pulling the median up? Clicking Animation / Micro reveals a wide distribution skewed right by two outlier films, while Horror / Low shows a tighter distribution. The same median can represent very different risk profiles.

Design decisions

Medians, not means. Movie revenue and ROI are heavily right-skewed. *Paranormal Activity* at 12,890× would dominate the Micro mean for Horror, making the average misleading for anyone who cannot replicate that specific outlier. The median is the value that half of films in a cell exceed; it is the honest “typical outcome” metric for a strategy.

What the data shows

Among cells with at least ten eligible films in the default cut:

- Animation / Micro: 7.5× median ROI, 89 % recovery
- Horror / Micro: 5.0× median ROI, 73 % recovery
- Horror / Low: 3.1× median ROI, 83 % recovery
- Horror / Blockbuster: 1.0× median ROI, 46 % recovery
- Drama / Low: 1.4× median ROI, 58 % recovery

The Horror row is the clearest single argument in the entire site for why scale matters within a genre. Horror at micro-budget is among the most efficient cells in the matrix; Horror at blockbuster scale barely recovers recorded budget. The genre is not inefficient — the scale is wrong.

Challenge: the \$10K floor and outlier sensitivity

We debated whether to raise the floor to \$50K or \$100K to reduce outlier sensitivity. Raising it would exclude *Paranormal Activity* entirely. We kept \$10K and use medians precisely *because* the outlier exists: it is real evidence that micro-budget films can achieve extreme returns, and the median correctly places it as one data point among 133 Horror / Micro films rather than distorting the summary.

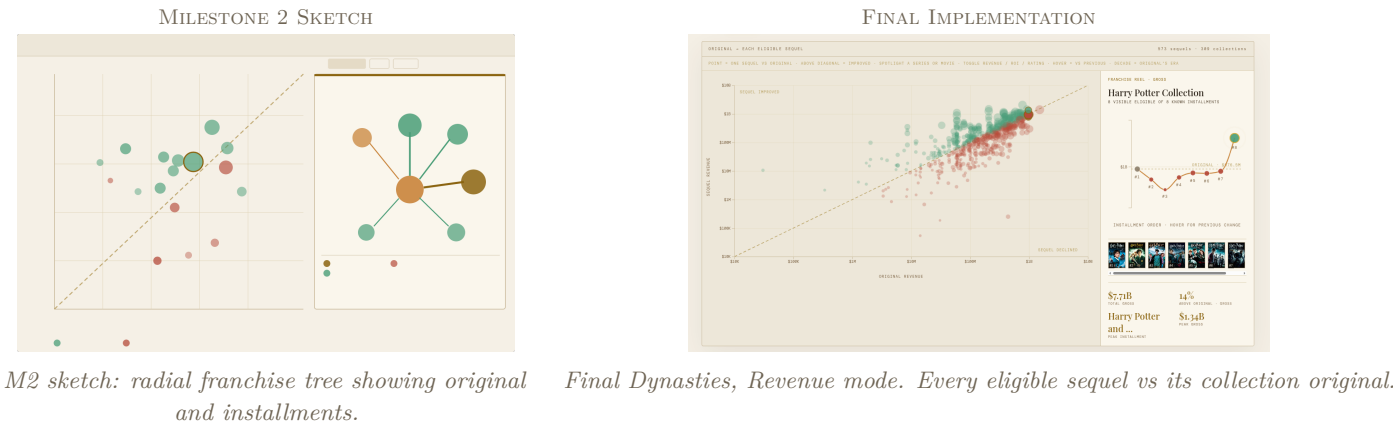
Matrix results: strongest and weakest cells, and decade context

Cell	<i>n</i>	Med. ROI	Recover
<i>Strongest by median ROI</i>			
Animation / Micro	27	7.50×	88.9 %
Horror / Micro	133	4.95×	72.9 %
Action / Micro	134	3.73×	82.1 %
Horror / Low	145	3.14×	83.4 %
<i>Weakest cells (≥10 films)</i>			
Horror / Block	13	0.97×	46.2 %
Thriller / Low	89	1.23×	52.8 %
Drama / Low	624	1.42×	58.3 %

Decade	<i>n</i>	Med. ROI	Recover	Med. gross
1970s	187	6.33×	91.4 %	\$26.5M
1980s	518	2.45×	71.2 %	\$21.6M
1990s	926	1.74×	64.6 %	\$27.3M
2000s	1,761	1.73×	67.4 %	\$39.0M
2010s	1,623	2.09×	70.1 %	\$38.7M

The 1970s pattern (6.33×) reflects a small eligible pool (187 films) rather than a universal rule about that era; the decade slider makes the sample size visible.

Dynasties: franchise inheritance over time



From sketch to final

The Milestone 2 sketch used a radial tree to show relational structure. The TA feedback singled out Dynasties as a view with strong potential, and in the revision process we identified a fundamental limitation of the radial design: it communicated connection but could not encode installment order, metric magnitude or three-way metric comparison. The final design replaces the radial tree with two coordinated components: a **all-eligible-sequel scatterplot** (original on x, sequel on y) and an **ordered franchise reel** (chronological line chart against the original baseline with poster strip). The reel opens when a user searches or selects a collection; it stays open while the Revenue/ROI/Rating toggle changes the metric, so the reader sees three different readings of the same franchise without any navigation. Another key design change: the M2 sketch compared only the **first sequel** with its original. The final version compares every eligible installment. Harry Potter has seven sequels, not one; reducing the franchise to “part two vs original” would discard six data points and miss the trajectory entirely.

What the data shows

Across 573 eligible sequel comparisons in 309 collections:

Measure	Improved	Share
Revenue	267 / 573	46.6 %
ROI	103 / 573	18.0 %
Rating (549 pairs)	76 / 549	13.8 %

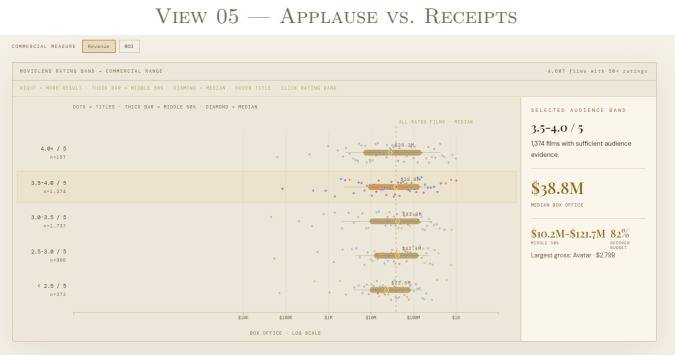
The Harry Potter Collection is the clearest three-lens demonstration. 1 of 7 eligible sequels outgrosses the original. 2 of 7 improve ROI. 6 of 7 improve audience rating. Audience memory and affection grew across the series; financial comparison against the opening film weakened on both gross and ROI. A franchise can be loved and commercially outrun by its own origin simultaneously. The gap between 46.6 % (gross) and 18.0 % (ROI) is the chapter’s central finding. Sequels routinely cost more to produce than the original, so expanding gross does not preserve return efficiency. Franchise expansion and franchise efficiency are different success stories.

Notable franchise collections — Revenue, ROI and Rating lenses

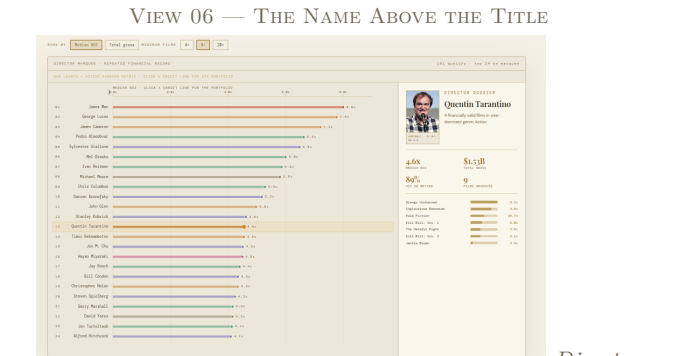
Collection	Revenue	ROI	Rating	Reading
Harry Potter	1 / 7	2 / 7	6 / 7	Audience approval grew across the series even as gross and ROI comparisons against the original weakened. The clearest three-lens example.
Star Wars	4 / 7	0 / 7	1 / 7	Large event grosses do not translate into better return efficiency or audience approval on the MovieLens evidence.
Fast & Furious	6 / 7	1 / 7	5 / 7	Gross expansion and approval improvement co-exist with ROI decline across nearly all installments.
The Dark Knight	2 / 2	2 / 2	2 / 2	Rare collection that improves on all three lenses. Shown as the exception, not the typical franchise outcome.
Toy Story	2 / 2	0 / 2	1 / 2	Later installments expand gross but do not improve return efficiency.

Figures show sequels that meet or beat the collection original on the selected measure, out of all eligible installments in the default cut.

Audience approval and filmmaker reputation



Rating bands, medians (◇), middle 50 % and film dots vs. all-rated baseline. Toggle Revenue/ROI.



Director portfolios ranked after selecting a minimum-film threshold. Side dossier shows individual films.

Applause: design and findings

View 05 uses **rating bands** rather than individual data points on a scatter to avoid implying that each rating difference is meaningful. MovieLens ratings cluster between 2.5 and 4.0; plotting 4,607 individual points would create a vertical cloud. The band design groups films by rating range and shows the commercial distribution *within* each group using a strip plot, thick the bar and diamond median. The design deliberately avoids a trend line. The central story is **overlap**: every rating band’s commercial range substantially overlaps every other. A trend line would suggest a cleaner relationship than exists.

MovieLens rating versus logged gross: $r=0.017$ so practically zero. The highest-rated band (4.0+) records \$28.3M median gross, *less* than the 3.0–3.5 band at \$42.9M. But 4.0+ reaches $4.99\times$ median ROI and 94.3 % budget recovery. Audience approval predicts return efficiency better than it predicts scale.

Challenge: two ratings systems

The dataset contains both TMDb vote averages (0–10 scale) and MovieLens ratings (0.5–5 scale). They correlate at $r\approx0.55$ but measure different audiences. We used only MovieLens for this view because it comes with a clear rating count, enabling the 50-rating evidence floor. TMDb vote averages have no accessible per-film count in the Kaggle export.

Directors: design and findings

View 06 introduces three evidence thresholds (4+, 6+, 10+ eligible films) before a director enters the ranking. This prevents a single outlier release from generating a misleading top position. The user selects the threshold and ranking metric (Median ROI or Total Gross), making the decision criteria explicit. Clicking a bar opens a portfolio dossier with all eligible films listed individually.

At the 6+ threshold in the default cut:

- James Wan: $8.0\times$ median ROI, \$2.38B total, 7 films
- Steven Spielberg: $4.2\times$ median ROI, \$9.26B total, 30 films

The two ranking metrics identify different directors as “most successful”. Efficiency and scale are genuinely different definitions of reputation.

How director portraits were obtained

The `fetch_director_portraits.py` script queries the Wikimedia Commons MediaWiki API with each director’s name. It retrieves the highest-resolution freely licensed portrait thumbnail, downloads it, converts to JPEG at 120px width and stores the source URL, licence identifier and author credit in `data/web/director_portraits.json`. Portraits display only in the expanded dossier panel, with a link to the Commons source.

Applause — full rating-band results (4,607 rating-qualified films, default cut)

Band	<i>n</i>	Med. gross	Med. ROI	Recover	Note
<2.5	373	\$23.5M	1.50×	61.9 %	Lowest gross and ROI; only group below break-even median.
2.5–3.0	966	\$42.1M	1.75×	65.9 %	Larger gross than top band despite lower ratings.
3.0–3.5	1,737	\$42.9M	1.97×	69.9 %	Highest median gross of all five bands.
3.5–4.0	1,374	\$38.8M	3.12×	81.9 %	Large sample; strong recovery rate.
4.0+	157	\$28.3M	4.99×	94.3 %	Smallest sample; highest ROI and recovery by wide margin.

Rating versus logged gross: $r=0.017$. Rating versus logged ROI: $r\approx0.200$. Audience approval is a substantially better predictor of return efficiency than of box-office scale. The 3.0–3.5 band earns more median gross than the 4.0+ band, yet returns only $1.97\times$ against $4.99\times$. This is the sharpest evidence in the site that applause and receipts measure different things.

How the site was built

Architecture

The website is a modular static D3 application deployable directly on GitHub Pages. Page structure and chapter containers live in `index.html`. Styling and both palettes are in `css/styles.css` using CSS custom properties, so the dark-room theme switch is a single class toggle on `<body>` with no JS intervention in chart code.

Module	Responsibility
<code>js/app.js</code>	Story text, loading, navigation, theme, shared controls
<code>js/state.js</code>	Genre/decade state, publishes to all views
<code>js/utlis.js</code>	Formatting, sampling, tooltips, log guard
<code>js/viz/*.js</code>	One D3 module per visualisation
<code>scripts/*.py</code>	Rebuild web exports and media metadata

Key technical challenges

Log scale ROI guard. Films with recorded gross below budget produce ROI values between 0 and 1. On a log axis these are valid but easy to misplace. `utlis.js` applies a minimum plotting floor for values that round to zero in log space, with a footnote: “N reported grosses below \$10K are pinned to the baseline.” Computed summaries still use the actual values.

Franchise reel poster URLs. TMDb poster URLs expire. The script `refresh_movie_posters.py` re-fetches current URLs for the top 200 films by gross, top 200 by ROI, and all 573 eligible franchise installments. Posters load only inside open tooltips and reels; the scatter never requests poster data during its initial render.

Static deployment with pre-computed data. Computing ROI, medians, correlation and franchise comparisons in the browser on every filter change would require shipping the full dataset and re-running analysis on each interaction. Instead, `export_web_data.py` pre-computes all filter combinations and writes compact JSON files. The browser only reads, never computes. This keeps interaction latency under 50ms.

Peer assessment

Member	Primary contributions
Hamza Barrada	Data preparation and analysis pipeline; ROI definitions and eligibility choices; Dossier Board work; repository documentation and result verification.
Amer Lakrami	Flow and Applause chapters; shared controls and interaction logic; front-end integration, hover/search behavior and interface testing.
Ziad Chentouf	Matrix and Dynasties chapters; franchise comparison logic; screencast; numerical validation and interpretation of findings.
Salma El Yadouni	Directors chapter; narrative structure, visual identity and process-book design; writing, editing and synthesis of design decisions.
Whole team	All members contributed to ideation, sketch refinement, design critique, debugging, result interpretation, visual polishing and final review.

Three findings that survived every filter test

- **Small bets multiply harder.** Micro-budget films return $2.98\times$ median ROI versus $2.24\times$ for blockbusters. Animation / Micro ($7.5\times$) and Horror / Micro ($5.0\times$) are the strongest specific strategies.
- **Franchise expansion loses efficiency.** 46.6 % of sequels outgross their original; only 18.0 % improve ROI.
- **Applause and receipts diverge.** Rating versus logged gross: $r=0.017$. The highest-rated films are not the highest-grossing.

Honest limitations

Recorded budget excludes marketing and distribution costs (which often equal or exceed production spend). $ROI \geq 1\times$ is budget recovery, not accounting profit. The dataset ends in 2017; streaming-era economics are outside its scope. All findings describe *observed association* in this specific eligible sample, not causal effects of production decisions.

CLOSING STATEMENT

*A movie can buy scale, earn applause, or inherit a name.
None of them buys certainty.*